

Lagrangian Dynamic Formulation of a Planar Mechanism with a Four-Bar Linkage Mounted on a Rotating Base

Technical Report — Dynamics Modeling

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Abstract

This report presents a detailed derivation of the equations of motion for a planar mechanism consisting of a rigid rotating arm (Body 1) on which a parallelogram four-bar linkage is mounted. The mechanism is actuated by a revolute actuator at the base joint of Body 1 and a prismatic actuator connected between a fixed point on Body 1 and the centre of mass of one side link. An external load is applied at a specified point on the coupler link. Using the Lagrangian formulation, the complete dynamic model in the standard manipulator form $M(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \boldsymbol{\tau} + \mathbf{Q}$ is derived in minimal coordinates $\mathbf{q} = [q_1, q_2]^\top$. The parallelogram constraint reduces the internal four-bar kinematics to two closed-form relations— $\alpha(q_2) = \text{const}$ and $\phi(q_2) = q_2$ —which substantially simplifies all dynamic quantities and yields a mass matrix and gravity vector structurally identical to those of a two-link serial planar arm.

1 Introduction

Four-bar mechanisms are widely used in robotic and mechatronic systems because they offer favourable force transmission and kinematic properties compared to serial chains. When such a mechanism is mounted on a moving rigid base, the dynamics must account for both the internal motion of the four-bar and the motion of the base itself.

In the mechanism considered here, Body 1 is a rigid arm that rotates about a fixed ground pivot O under the action of a revolute actuator. At its distal end, a parallelogram four-bar linkage is rigidly attached. The parallelogram configuration—where the two side links are equal in length and remain parallel throughout the entire range of motion—eliminates the nonlinear algebraic constraint that normally characterises four-bar mechanisms and allows all dependent angles to be expressed in closed form as simple functions of the single independent joint coordinate q_2 .

The second actuator is a prismatic (linear) joint assumed to be massless and to have no rotational inertia because we use cable to transmit effort at two sides. It connects a fixed point B_p on Body 1 to the centre of mass G_3 of the upper side link. Because both endpoints rotate together with the mechanism, the actuator length L_p depends only on q_2 and not on q_1 .

An external varying load force $\mathbf{F}$ is applied at a specified point G_4 on the coupler link. This formulation follows the methodology of Tang [1], extended to a 2-DOF system with a prismatic actuator and a moving base.

2 System Description and Notation

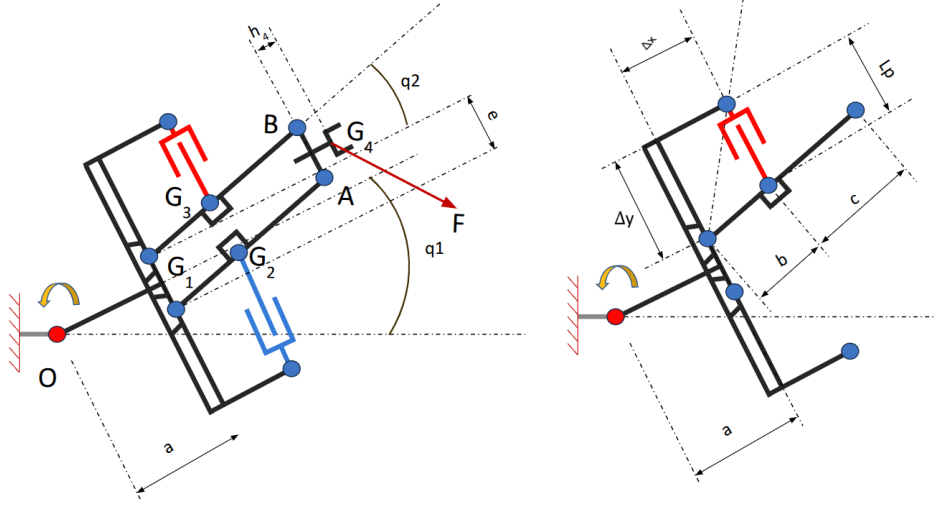


Figure 1: Robot structure

2.1 Mechanism Components

The mechanism consists of four moving bodies.

- **body 1 (Body 1):** A rigid arm that rotates about ground pivot O . Its centre of mass G_1 lies at distance a from O along the body axis. Points O_2 (lower) and O_3 (upper) are rigidly attached to Body 1 and serve as the two ground passive pivots of the four-bar linkage, with G_1 the midpoint of segment O_2O_3 .
- **body 2:** A rigid link of length $L = b + c$, pivoting at O_2 with its centre of mass G_2 at distance b from O_2 along the link axis.
- **body 3:** A rigid link of length $L = b + c$, pivoting at O_3 , identical in mass distribution to body 2, with centre of mass G_3 at distance b from O_3 . The prismatic actuator connects to G_3 .
- **body 4 (Coupler):** A rigid body connecting the distal end of body 2 and end of body 3. The point G_4 , which serves as the location of the external load, lies on the perpendicular bisector of segment AB at a perpendicular distance h_4 from the midpoint M of AB . The coupler carries mass m_4 and moment of inertia I_4 .

2.2 Parameters

2.3 Generalized Coordinates

The system has two degrees of freedom parameterised by

$$\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}, \quad (1)$$

where q_1 is the absolute angle of Body 1 measured from the positive x -axis of the inertial frame, and q_2 is the angle of the side links measured from the axis of Body 1 (a relative

Table 1: Relevant parameters for the 2-DOF mechanism under consideration.

Symbol	Definition
a	Distance from O to G_1 (Body 1 CoM); also distance from O to midpoint of O_2O_3
e	Length of segment O_2O_3 (= coupler length AB)
b	Distance from pivot to CoM along each side link
$L = b + c$	Total length of each side link ($l_1 = l_3 = b + c$)
h_4	Perpendicular offset from midpoint of AB to load point G_4
x_B, y_B	Coordinates of prismatic endpoint B_p in the body-1 frame
m_1, I_1	Mass and moment of inertia of Body 1 about its centre of mass G_1
m_s, I_s	Mass and moment of inertia of each side link about its own CoM ($m_2 = m_3 = m_s, I_2 = I_3 = I_s$)
m_4, I_4	Mass and moment of inertia of the coupler about G_4
g	Gravitational acceleration

angle within the body-1 frame). The absolute angle of each side link in the inertial frame is therefore $q_1 + q_2$.

3 Kinematic Analysis

3.1 Reference Frames and Coordinate Setup

The inertial frame has its origin at O , with the x -axis horizontal and y -axis vertical. The body-1 frame is attached to Body 1, with its origin at O , x -axis directed along the body axis, and y -axis perpendicular. The transformation from the body-1 frame to the inertial frame is given by the rotation matrix

$$R(q_1) = \begin{bmatrix} \cos q_1 & -\sin q_1 \\ \sin q_1 & \cos q_1 \end{bmatrix}. \quad (2)$$

Any point with position $\mathbf{r}^B = (x^B, y^B)^\top$ in the body-1 frame has inertial position

$$\mathbf{r}^I = R(q_1) \mathbf{r}^B. \quad (3)$$

3.2 Positions of Key Points in the Body-1 Frame

The positions below are fixed in the body-1 frame (functions of q_2 only) and follow directly from the geometry described in Section 2:

$$\mathbf{O}_2 = \begin{pmatrix} a \\ -e/2 \end{pmatrix}, \quad \mathbf{O}_3 = \begin{pmatrix} a \\ +e/2 \end{pmatrix}, \quad \mathbf{G}_1 = \begin{pmatrix} a \\ 0 \end{pmatrix} = \frac{\mathbf{O}_2 + \mathbf{O}_3}{2}. \quad (4)$$

The centres of mass of the two side links are

$$\mathbf{G}_2(q_2) = \begin{pmatrix} a + b \cos q_2 \\ -e/2 + b \sin q_2 \end{pmatrix}, \quad \mathbf{G}_3(q_2) = \begin{pmatrix} a + b \cos q_2 \\ +e/2 + b \sin q_2 \end{pmatrix}. \quad (5)$$

The distal ends of the side links and the midpoint of the coupler are

$$\mathbf{A}(q_2) = \begin{pmatrix} a + L \cos q_2 \\ -e/2 + L \sin q_2 \end{pmatrix}, \quad \mathbf{B}(q_2) = \begin{pmatrix} a + L \cos q_2 \\ +e/2 + L \sin q_2 \end{pmatrix}, \quad (6)$$

$$\mathbf{G}_4(q_2) = \begin{pmatrix} a + L \cos q_2 + h_4 \\ L \sin q_2 \end{pmatrix}, \quad (7)$$

3.3 Parallelogram Constraint and Simplification

The mechanism satisfies the loop-closure equation

$$\mathbf{O}_3 + L \begin{pmatrix} \cos \theta_2 \\ \sin \theta_2 \end{pmatrix} = \mathbf{O}_2 + L \begin{pmatrix} \cos \theta_3 \\ \sin \theta_3 \end{pmatrix} + \begin{pmatrix} 0 \\ e \end{pmatrix}, \quad (8)$$

where θ_2, θ_3 are the absolute angles of bodies 2 and 3 in the inertial frame. Because $l_1 = l_3 = L$ and O_2O_3 is parallel to AB (both vertical in the body-1 frame, both of length e), this is a parallelogram mechanism. The constraint is satisfied for all q_2 if and only if

$$\theta_3 = \theta_2 = q_1 + q_2, \quad (9)$$

and the coupler angle in the body-1 frame is

$$\alpha(q_2) = \text{const.} \quad (10)$$

In Tang's notation, the velocity ratios are

$$S_1 = \frac{\partial \alpha}{\partial q_2} = 0, \quad S_2 = \frac{\partial \phi}{\partial q_2} = 1. \quad (11)$$

These are the only two velocity ratios needed to parameterise the entire four-bar internal kinematics, and both take constant values owing to the parallelogram geometry.

3.4 Angular Velocities

From (9) and (10), the absolute angular velocities of each body are:

Body 1:	$\omega_1 = \dot{q}_1,$
body 2:	$\omega_2 = \dot{q}_1 + \dot{q}_2,$
body 3:	$\omega_3 = \dot{q}_1 + \dot{q}_2,$
body 4 (coupler, does not rotate in body-1 frame):	$\omega_4 = \dot{q}_1.$

3.5 Velocities of the Centres of Mass

For any point with body-1 frame coordinates $\mathbf{r}^B(q_2)$, the inertial velocity is obtained by differentiating (3):

$$\dot{\mathbf{r}}^I = \dot{q}_1 R(q_1) \begin{pmatrix} -y^B \\ x^B \end{pmatrix} + R(q_1) \frac{d\mathbf{r}^B}{dq_2} \dot{q}_2. \quad (12)$$

Since $R(q_1)$ is orthogonal, the squared inertial speed is

$$\|\dot{\mathbf{r}}^I\|^2 = \|\dot{\mathbf{r}}^B\|^2 + (x^{B2} + y^{B2})\dot{q}_1^2 + 2\dot{q}_1 (x^B \dot{y}^B - y^B \dot{x}^B), \quad (13)$$

where $\dot{x}^B = (\partial x^B / \partial q_2) \dot{q}_2$ and similarly for $\dot{y}^B$.

Applying (13) to each centre of mass:

Body 1. G_1 is fixed in the body-1 frame, so $\dot{\mathbf{G}}_1^B = \mathbf{0}$. Defining $r_1^2 := a^2$,

$$\|\dot{\mathbf{r}}_{G_1}^I\|^2 = a^2 \dot{q}_1^2. \quad (14)$$

Body 2. $G_2^B = (a + bc_2, -e/2 + bs_2)$:

$$\begin{aligned} \dot{\mathbf{G}}_2^B &= (-b \sin q_2, b \cos q_2) \dot{q}_2, & \|\dot{\mathbf{G}}_2^B\|^2 &= b^2 \dot{q}_2^2, \\ r_2^2 &= a^2 + 2ab \cos q_2 + b^2 + \frac{e^2}{4} - eb \sin q_2. \\ x_{G_2}^B \dot{y}_{G_2}^B - y_{G_2}^B \dot{x}_{G_2}^B &= b \dot{q}_2 \left[a \cos q_2 - \frac{e}{2} \sin q_2 + b \right], \end{aligned} \quad (15)$$

$$\|\dot{\mathbf{r}}_{G_2}^I\|^2 = b^2 \dot{q}_2^2 + r_2^2 \dot{q}_1^2 + 2b \dot{q}_1 \dot{q}_2 \left[a \cos q_2 - \frac{e}{2} \sin q_2 + b \right]. \quad (16)$$

Body 3. $G_3^B = (a + bc_2, +e/2 + bs_2)$:

$$\begin{aligned} \dot{\mathbf{G}}_3^B &= (-b \sin q_2, b \cos q_2) \dot{q}_2, & \|\dot{\mathbf{G}}_3^B\|^2 &= b^2 \dot{q}_2^2, \\ r_3^2 &= a^2 + 2ab \cos q_2 + b^2 + \frac{e^2}{4} + eb \sin q_2. \\ x_{G_3}^B \dot{y}_{G_3}^B - y_{G_3}^B \dot{x}_{G_3}^B &= b \dot{q}_2 \left[a \cos q_2 + \frac{e}{2} \sin q_2 + b \right], \end{aligned} \quad (17)$$

$$\|\dot{\mathbf{r}}_{G_3}^I\|^2 = b^2 \dot{q}_2^2 + r_3^2 \dot{q}_1^2 + 2b \dot{q}_1 \dot{q}_2 \left[a \cos q_2 + \frac{e}{2} \sin q_2 + b \right]. \quad (18)$$

Body 4 (coupler). $G_4^B = (a + Lc_2 + h_4, Ls_2)$:

$$\begin{aligned} \dot{\mathbf{G}}_4^B &= (-L \sin q_2, L \cos q_2) \dot{q}_2, & \|\dot{\mathbf{G}}_4^B\|^2 &= L^2 \dot{q}_2^2, \\ r_4^2 &= (a + h_4)^2 + 2(a + h_4)L \cos q_2 + L^2. \\ x_{G_4}^B \dot{y}_{G_4}^B - y_{G_4}^B \dot{x}_{G_4}^B &= L \dot{q}_2 [(a + h_4) \cos q_2 + L], \end{aligned} \quad (19)$$

$$\|\dot{\mathbf{r}}_{G_4}^I\|^2 = L^2 \dot{q}_2^2 + r_4^2 \dot{q}_1^2 + 2L \dot{q}_1 \dot{q}_2 [(a + h_4) \cos q_2 + L]. \quad (20)$$

4 Lagrangian Formulation

4.1 Kinetic Energy

The total kinetic energy is $T = T_1 + T_2 + T_3 + T_4$, where $T_i = \frac{1}{2}m_i\|\dot{\mathbf{r}}_{G_i}^I\|^2 + \frac{1}{2}I_i\omega_i^2$.

Body 1:

$$T_1 = \frac{1}{2}(m_1a^2 + I_1)\dot{q}_1^2. \quad (21)$$

body 2:

$$\begin{aligned} T_2 = & \frac{1}{2}(m_sb^2 + I_s)\dot{q}_2^2 + \frac{1}{2}(m_sr_2^2 + I_s)\dot{q}_1^2 \\ & + \left[m_sb \left(a \cos q_2 - \frac{e}{2} \sin q_2 + b \right) + I_s \right] \dot{q}_1 \dot{q}_2. \end{aligned} \quad (22)$$

body 3:

$$\begin{aligned} T_3 = & \frac{1}{2}(m_sb^2 + I_s)\dot{q}_2^2 + \frac{1}{2}(m_sr_3^2 + I_s)\dot{q}_1^2 \\ & + \left[m_sb \left(a \cos q_2 + \frac{e}{2} \sin q_2 + b \right) + I_s \right] \dot{q}_1 \dot{q}_2. \end{aligned} \quad (23)$$

body 4:

$$\begin{aligned} T_4 = & \frac{1}{2}m_4L^2\dot{q}_2^2 + \frac{1}{2}(m_4r_4^2 + I_4)\dot{q}_1^2 \\ & + m_4L \left[(a + h_4) \cos q_2 + L \right] \dot{q}_1 \dot{q}_2. \end{aligned} \quad (24)$$

body 4's inertia contributes $I_4\dot{q}_1^2$ but no $I_4\dot{q}_2^2$ term, which is a direct consequence of the coupler having zero angular velocity in the body-1 frame (Eq. 10).

4.2 Total Kinetic Energy and Mass Matrix

Summing T_1 through T_4 , the total kinetic energy takes the standard form

$$T = \frac{1}{2}\dot{\mathbf{q}}^\top M(q_2) \dot{\mathbf{q}}, \quad (25)$$

where $M(q_2)$ is a symmetric positive-definite 2×2 matrix depending only on q_2 . Define the composite constants

$$K_1 := 2m_sab + m_4(a + h_4)L, \quad (26)$$

$$K_2 := 2m_sb^2 + 2I_s + m_4L^2, \quad (27)$$

$$M_{11,0} := m_1a^2 + I_1 + 2m_s \left(a^2 + b^2 + \frac{e^2}{4} \right) + 2I_s + m_4 \left[(a + h_4)^2 + L^2 \right] + I_4. \quad (28)$$

The mass matrix entries are then

$$M_{11}(q_2) = M_{11,0} + 2K_1 \cos q_2, \quad (29)$$

$$M_{12}(q_2) = K_2 + K_1 \cos q_2, \quad (30)$$

$$M_{22} = K_2 \quad (\text{constant}). \quad (31)$$

Several observations follow directly from (29)–(31):

1. M_{22} is *constant* in q_2 . The coupler does not rotate relative to Body 1, so it contributes no q_2 -dependent rotational inertia.
2. The partial derivatives satisfy $M'_{11} = 2M'_{12}$, where primes denote d/dq_2 . Specifically,

$$M'_{11}(q_2) = -2K_1 \sin q_2, \quad M'_{12}(q_2) = -K_1 \sin q_2. \quad (32)$$

3. The minimum eigenvalue of M is bounded below by $K_2 > 0$ for all q_2 , ensuring the system is never dynamically singular.

4.3 Potential Energy

Taking the y -axis of the inertial frame as the vertical direction (positive upward), the gravitational potential energy is $V_i = m_i g y_{G_i}^I$ where $y^I = x^B \sin q_1 + y^B \cos q_1$. Evaluating at each centre of mass and summing, with the symmetry $m_2 = m_3 = m_s$ causing the $\pm(e/2) \sin q_2$ terms to cancel,

$$V(q_1, q_2) = g \left[\underbrace{((m_1 + 2m_s + m_4)a + m_4 h_4)}_{m_{\text{eff},1}} \sin q_1 + \underbrace{(2m_s b + m_4 L)}_{m_{\text{eff},2}} \sin(q_1 + q_2) \right], \quad (33)$$

where the two effective mass parameters are defined as

$$m_{\text{eff},1} := (m_1 + 2m_s + m_4)a + m_4 h_4, \quad (34)$$

$$m_{\text{eff},2} := 2m_s b + m_4 L. \quad (35)$$

The potential energy (33) is structurally identical to that of a two-link planar serial robot. This equivalence is a direct consequence of the parallelogram geometry.

4.4 Lagrangian

The Lagrangian of the complete system is $\mathcal{L} = T - V$, which gives

$$\mathcal{L} = \frac{1}{2} [M_{11} \dot{q}_1^2 + 2M_{12} \dot{q}_1 \dot{q}_2 + K_2 \dot{q}_2^2] - g [m_{\text{eff},1} \sin q_1 + m_{\text{eff},2} \sin(q_1 + q_2)]. \quad (36)$$

5 Equations of Motion

5.1 Derivation via Euler–Lagrange Equations

The Euler–Lagrange equations for the unconstrained coordinates q_1 and q_2 are

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = \mathcal{Q}_i, \quad i = 1, 2, \quad (37)$$

where $\mathcal{Q}_i$ are the generalised forces, derived in Section 6.

The momenta conjugate to q_1 and q_2 are

$$\frac{\partial \mathcal{L}}{\partial \dot{q}_1} = M_{11}(q_2) \dot{q}_1 + M_{12}(q_2) \dot{q}_2, \quad (38)$$

$$\frac{\partial \mathcal{L}}{\partial \dot{q}_2} = M_{12}(q_2) \dot{q}_1 + K_2 \dot{q}_2. \quad (39)$$

Differentiating (38) and (39) with respect to time:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_1} \right) = M_{11} \ddot{q}_1 + M_{12} \ddot{q}_2 + M'_{11} \dot{q}_1 \dot{q}_2 + M'_{12} \dot{q}_2^2, \quad (40)$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_2} \right) = M_{12} \ddot{q}_1 + K_2 \ddot{q}_2 + M'_{12} \dot{q}_1 \dot{q}_2. \quad (41)$$

The partial derivatives of $\mathcal{L}$ with respect to the coordinates are

$$\frac{\partial \mathcal{L}}{\partial q_1} = -g[m_{\text{eff},1} \cos q_1 + m_{\text{eff},2} \cos(q_1 + q_2)], \quad (42)$$

$$\frac{\partial \mathcal{L}}{\partial q_2} = \frac{1}{2} M'_{11} \dot{q}_1^2 + M'_{12} \dot{q}_1 \dot{q}_2 - g m_{\text{eff},2} \cos(q_1 + q_2). \quad (43)$$

Note that $\partial \mathcal{L} / \partial q_1$ does not contain any velocity-squared terms because T is independent of q_1 (M depends only on q_2).

Substituting (40) and (42) into (37) for $i = 1$:

$$M_{11} \ddot{q}_1 + M_{12} \ddot{q}_2 + M'_{11} \dot{q}_1 \dot{q}_2 + M'_{12} \dot{q}_2^2 + g[m_{\text{eff},1} \cos q_1 + m_{\text{eff},2} \cos(q_1 + q_2)] = \mathcal{Q}_1. \quad (44)$$

Substituting (41) and (43) into (37) for $i = 2$ (the $M'_{12} \dot{q}_1 \dot{q}_2$ terms cancel):

$$M_{12} \ddot{q}_1 + K_2 \ddot{q}_2 + K_1 \sin q_2 \dot{q}_1^2 + g m_{\text{eff},2} \cos(q_1 + q_2) = \mathcal{Q}_2, \quad (45)$$

where $-\frac{1}{2} M'_{11} = K_1 \sin q_2$ from (32).

5.2 Coriolis and Centrifugal Matrix

The Coriolis and centrifugal matrix $C(q_2, \dot{\mathbf{q}})$ is defined via the Christoffel symbols of the first kind

$$c_{ijk} = \frac{1}{2} \left(\frac{\partial M_{ij}}{\partial q_k} + \frac{\partial M_{ik}}{\partial q_j} - \frac{\partial M_{jk}}{\partial q_i} \right). \quad (46)$$

Since M depends only on q_2 , all partial derivatives with respect to q_1 vanish identically. Computing the non-zero Christoffel symbols:

$$c_{112} = c_{121} = \frac{1}{2} M'_{11} = -K_1 \sin q_2, \quad (47)$$

$$c_{122} = M'_{12} = -K_1 \sin q_2, \quad (48)$$

$$c_{211} = -\frac{1}{2} M'_{11} = K_1 \sin q_2, \quad (49)$$

$$c_{212} = c_{221} = c_{222} = c_{111} = 0. \quad (50)$$

The Coriolis matrix $C_{ij} = \sum_k c_{ijk} \dot{q}_k$ is

$$C(q_2, \dot{\mathbf{q}}) = K_1 \sin q_2 \begin{bmatrix} -\dot{q}_2 & -(\dot{q}_1 + \dot{q}_2) \\ \dot{q}_1 & 0 \end{bmatrix}. \quad (51)$$

One can verify that $\dot{M} - 2C$ is skew-symmetric—a standard property of Lagrangian dynamics—confirming (51). In particular,

$$\dot{M}_{11} = M'_{11} \dot{q}_2 = -2K_1 \sin q_2 \dot{q}_2, \quad 2C_{11} = -2K_1 \sin q_2 \dot{q}_2 \quad \Rightarrow \quad \dot{M}_{11} - 2C_{11} = 0. \quad (52)$$

The Coriolis and centrifugal term $C\dot{\mathbf{q}}$ evaluates to

$$(C\dot{\mathbf{q}})_1 = -K_1 \sin q_2 (2\dot{q}_1 \dot{q}_2 + \dot{q}_2^2), \quad (53)$$

$$(C\dot{\mathbf{q}})_2 = +K_1 \sin q_2 \dot{q}_1^2. \quad (54)$$

5.3 Gravity Vector

The gravity vector $\mathbf{G}(\mathbf{q}) = \nabla_{\mathbf{q}} V$ has components

$$G_1(\mathbf{q}) = \frac{\partial V}{\partial q_1} = g[m_{\text{eff},1} \cos q_1 + m_{\text{eff},2} \cos(q_1 + q_2)], \quad (55)$$

$$G_2(\mathbf{q}) = \frac{\partial V}{\partial q_2} = g m_{\text{eff},2} \cos(q_1 + q_2). \quad (56)$$

Note that G_2 does not depend on q_1 independently—only through the combination $(q_1 + q_2)$, which is the absolute orientation of the side links in the inertial frame.

6 Generalised Forces

6.1 Actuator 1 — Revolute Joint at O

Actuator 1 applies a torque τ_1 directly at the pivot O of Body 1. This is a joint actuator for coordinate q_1 , so

$$\boldsymbol{\tau}_1 = \begin{bmatrix} \tau_1 \\ 0 \end{bmatrix}. \quad (57)$$

6.2 Actuator 2 — Prismatic Joint

Actuator 2 is a massless, inertia-free prismatic actuator connecting a fixed point B_p in the body-1 frame to the centre of mass G_3 of Part 3. Let

$$\mathbf{B}_p = \begin{pmatrix} x_B \\ y_B \end{pmatrix}, \quad \mathbf{G}_3(q_2) = \begin{pmatrix} a + b \cos q_2 \\ +e/2 + b \sin q_2 \end{pmatrix}. \quad (58)$$

Define the offsets (note: G_3 is the upper link, so $y_B - e/2$ appears):

$$\Delta x := x_B - a, \quad \Delta y := y_B - \frac{e}{2}. \quad (59)$$

The length of the prismatic actuator is

$$\begin{aligned} L_p^2(q_2) &= (\Delta x - b \cos q_2)^2 + (\Delta y - b \sin q_2)^2 \\ &= \Delta x^2 + \Delta y^2 + b^2 - 2b(\Delta x \cos q_2 + \Delta y \sin q_2). \end{aligned} \quad (60)$$

Since both B_p and G_3 are defined in the body-1 frame, L_p depends only on q_2 and not on q_1 . Differentiating (60):

$$\frac{\partial L_p}{\partial q_2} = \frac{b(\Delta x \sin q_2 - \Delta y \cos q_2)}{L_p(q_2)}. \quad (61)$$

The rate of change of actuator length is $\dot{L}_p = (\partial L_p / \partial q_2) \dot{q}_2$. The virtual work of the prismatic force f_p (positive extending) is $\delta W_p = f_p \delta L_p = f_p (\partial L_p / \partial q_2) \delta q_2$. Therefore the generalised forces are

$$Q_{q_1}^{(\text{pris})} = 0, \quad Q_{q_2}^{(\text{pris})} = f_p \frac{\partial L_p}{\partial q_2} = \frac{f_p b(\Delta x \sin q_2 - \Delta y \cos q_2)}{L_p(q_2)}. \quad (62)$$

The vanishing of $Q_{q_1}^{(\text{pris})}$ confirms that the prismatic actuator exerts no net torque on the base coordinate q_1 .

A **singular configuration** of the prismatic actuator occurs when $\partial L_p / \partial q_2 = 0$. Setting the numerator of (61) to zero:

$$\Delta x \sin q_2 - \Delta y \cos q_2 = 0 \implies \tan q_2^* = \frac{\Delta y}{\Delta x}, \quad (63)$$

$$\boxed{q_2^* = \text{atan2}(\Delta y, \Delta x) \quad \text{and} \quad q_2^* + \pi.} \quad (64)$$

There are two singular configurations per period; both must be excluded from the operational workspace.

If passive viscous damping of coefficient $d > 0$ acts at the prismatic joint, an additional generalised force

$$Q_{q_2}^{(\text{damp})} = -d \dot{L}_p \frac{\partial L_p}{\partial q_2} = -d \left(\frac{\partial L_p}{\partial q_2} \right)^2 \dot{q}_2 \quad (65)$$

is added to the right-hand side of the q_2 equation. This represents a configuration-dependent viscous damping term.

6.3 External Load at G_4

An external force $\mathbf{F} = (F_x, F_y)^\top$ is applied at G_4 in the inertial frame. The generalised forces are computed via the Jacobian $J_{G_4}(\mathbf{q}) = \partial \mathbf{r}_{G_4}^I / \partial \mathbf{q}$. The two columns, obtained from (12), are

$$\frac{\partial \mathbf{r}_{G_4}^I}{\partial q_1} = R(q_1) \begin{pmatrix} -L \sin q_2 \\ a + L \cos q_2 + h_4 \end{pmatrix}, \quad (66)$$

$$\frac{\partial \mathbf{r}_{G_4}^I}{\partial q_2} = R(q_1) \begin{pmatrix} -L \sin q_2 \\ L \cos q_2 \end{pmatrix} = \begin{pmatrix} -L \sin(q_1 + q_2) \\ L \cos(q_1 + q_2) \end{pmatrix}. \quad (67)$$

The generalised forces from the external load are $\mathbf{Q}_F = J_{G_4}^\top \mathbf{F}$:

$$(Q_F)_1 = -L s_2 (F_x c_1 + F_y s_1) + (a + L c_2 + h_4) (F_y c_1 - F_x s_1), \quad (68)$$

$$(Q_F)_2 = L [F_y \cos(q_1 + q_2) - F_x \sin(q_1 + q_2)], \quad (69)$$

where $c_1 = \cos q_1$, $s_1 = \sin q_1$, $c_2 = \cos q_2$, $s_2 = \sin q_2$.

Special case. If $\mathbf{F} = (0, -m_F g)^\top$ (gravity only, load mass m_F at G_4), equations (68)–(69) are equivalently obtained by the substitution $m_4 \rightarrow m_4 + m_F$ in (34)–(35).

7 Complete Equations of Motion

Assembling the terms from Sections 5 and 6, the complete equations of motion in standard manipulator form are

$$\boxed{M(q_2) \ddot{\mathbf{q}} + C(q_2, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \boldsymbol{\tau}_{\text{act}} + \mathbf{Q}_F,} \quad (70)$$

where the individual matrices and vectors are

$$M(q_2) = \begin{bmatrix} M_{11,0} + 2K_1 \cos q_2 & K_2 + K_1 \cos q_2 \\ K_2 + K_1 \cos q_2 & K_2 \end{bmatrix}, \quad (71)$$

$$C(q_2, \dot{\mathbf{q}}) = K_1 \sin q_2 \begin{bmatrix} -\dot{q}_2 & -(\dot{q}_1 + \dot{q}_2) \\ \dot{q}_1 & 0 \end{bmatrix}, \quad (72)$$

$$\mathbf{G}(\mathbf{q}) = g \begin{bmatrix} m_{\text{eff},1} \cos q_1 + m_{\text{eff},2} \cos(q_1 + q_2) \\ m_{\text{eff},2} \cos(q_1 + q_2) \end{bmatrix}, \quad (73)$$

$$\boldsymbol{\tau}_{\text{act}} = \begin{bmatrix} \tau_1 \\ \frac{f_p b(\Delta x \sin q_2 - \Delta y \cos q_2)}{L_p(q_2)} \end{bmatrix}, \quad (74)$$

and $\mathbf{Q}_F$ is given by (68)–(69).

The explicit scalar equations are as follows.

Equation 1 (revolute actuator):

$$(M_{11,0} + 2K_1 c_2) \ddot{q}_1 + (K_2 + K_1 c_2) \ddot{q}_2 - K_1 s_2 (2\dot{q}_1 \dot{q}_2 + \dot{q}_2^2) + g[m_{\text{eff},1} c_1 + m_{\text{eff},2} c_{12}] = \tau_1 + (Q_F)_1. \quad (75)$$

Equation 2 (prismatic actuator):

$$(K_2 + K_1 c_2) \ddot{q}_1 + K_2 \ddot{q}_2 + K_1 s_2 \dot{q}_1^2 + g m_{\text{eff},2} c_{12} = \frac{f_p b(\Delta x s_2 - \Delta y c_2)}{L_p} + (Q_F)_2, \quad (76)$$

where $c_1 = \cos q_1$, $c_2 = \cos q_2$, $s_2 = \sin q_2$, $c_{12} = \cos(q_1 + q_2)$.

8 State-Space Representation and Forward Dynamics

Define the state vector $\mathbf{x} = [q_1, q_2, \dot{q}_1, \dot{q}_2]^\top$. The forward dynamics are obtained by solving (70) for $\ddot{\mathbf{q}}$:

$$\ddot{\mathbf{q}} = M^{-1}(q_2)[\boldsymbol{\tau}_{\text{act}} + \mathbf{Q}_F - C(q_2, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{G}(\mathbf{q})]. \quad (77)$$

The determinant of $M(q_2)$ is

$$\det M = (M_{11,0} + 2K_1 c_2) K_2 - (K_2 + K_1 c_2)^2 = K_2(M_{11,0} - K_2) - K_1^2 \cos^2 q_2. \quad (78)$$

Positive definiteness of M requires $\det M > 0$ for all q_2 . Since the minimum of (78) over q_2 is achieved at $\cos^2 q_2 = 1$, the necessary and sufficient condition is

$$K_2(M_{11,0} - K_2) > K_1^2. \quad (79)$$

This inequality should be verified for the specific numerical parameters of the mechanism before deployment.

9 Summary of Defined Constants

For reference, all composite constants appearing in the dynamic model are collected in Table 2.

Table 2: Composite constants in the dynamic model (equations labelled as in the main text).

Symbol	Definition
$K_1 = 2m_s ab + m_4(a + h_4)L$	Eq. (26)
$K_2 = 2m_s b^2 + 2I_s + m_4 L^2$	Eq. (27); also M_{22}
$M_{11,0}$	Eq. (28) (full expression)
$m_{\text{eff},1} = (m_1 + 2m_s + m_4)a + m_4 h_4$	Eq. (34)
$m_{\text{eff},2} = 2m_s b + m_4 L$	Eq. (35)
$\Delta x = x_B - a, \quad \Delta y = y_B - e/2$	Eq. (59)

10 Special Properties and Remarks

10.1 Equivalence to a 2-link serial arm. The mass matrix (71) and gravity vector (73) are structurally identical to those of a planar 2R serial manipulator with link-1 effective mass $m_{\text{eff},1}$ and link-2 effective mass $m_{\text{eff},2}$. This equivalence, which arises entirely from the parallelogram constraint (11), implies that: (a) all control strategies designed for 2R serial arms—including computed-torque control, PD-plus-gravity compensation, and adaptive control—can be applied directly with no modification; (b) the mechanism can be simulated as a 2R arm with modified inertia parameters, avoiding the computational overhead of enforcing closed-chain constraints.

10.2 M depends only on q_2 . Because M does not depend on q_1 , the inertia seen by Actuator 1 depends only on the internal configuration q_2 of the four-bar, not on the absolute orientation of Body 1.

10.3 Prismatic actuator singularity. The prismatic actuator cannot actuate q_2 at $q_2^* = \text{atan2}(\Delta y, \Delta x)$ and $q_2^* + \pi$ (Eq. 64). The operational workspace must exclude a neighbourhood around both values. This is analogous to a robot wrist singularity.

10.4 Asymmetric mass case. If $m_2 \neq m_3$ (e.g. due to added hardware on one side link), the potential energy gains an extra term $g(m_2 - m_3)(e/2) \sin q_1$ contributing to G_1 , and M_{11} gains a term $(m_2 - m_3)eb \sin q_2$. The entry M_{12} gains $(m_2 - m_3)b(e/2) \sin q_2$, and the structural equivalence to the symmetric case is broken.

11 Conclusion

A complete Lagrangian dynamic model for the parallelogram four-bar mechanism mounted on a rotating base has been derived. The system possesses two degrees of freedom parameterised by the base angle q_1 and the relative crank angle q_2 . The parallelogram constraint reduces the four-bar internal kinematics to the trivial relations $S_1 = 0$ and $S_2 = 1$, allowing all kinematic and dynamic quantities to be expressed in closed form without any iterative solution.

The mass matrix $M(q_2)$ has a simple trigonometric structure parameterised by two scalar constants K_1 and K_2 , with $M_{22} = K_2$ constant (Eqs. 26–31). The Coriolis matrix is factored by $K_1 \sin q_2$ (Eq. 72), and the gravity vector reduces to the standard two-link arm form in terms of $m_{\text{eff},1}$ and $m_{\text{eff},2}$ (Eqs. 34–35). The generalised force from the massless prismatic actuator is given analytically in Eq. (62), and its moment arm $\partial L_p / \partial q_2$ (Eq. 61) provides a direct measure of actuator effectiveness as a function of configuration. The complete equations of motion are given in scalar form in Eqs. (75)–(76).

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